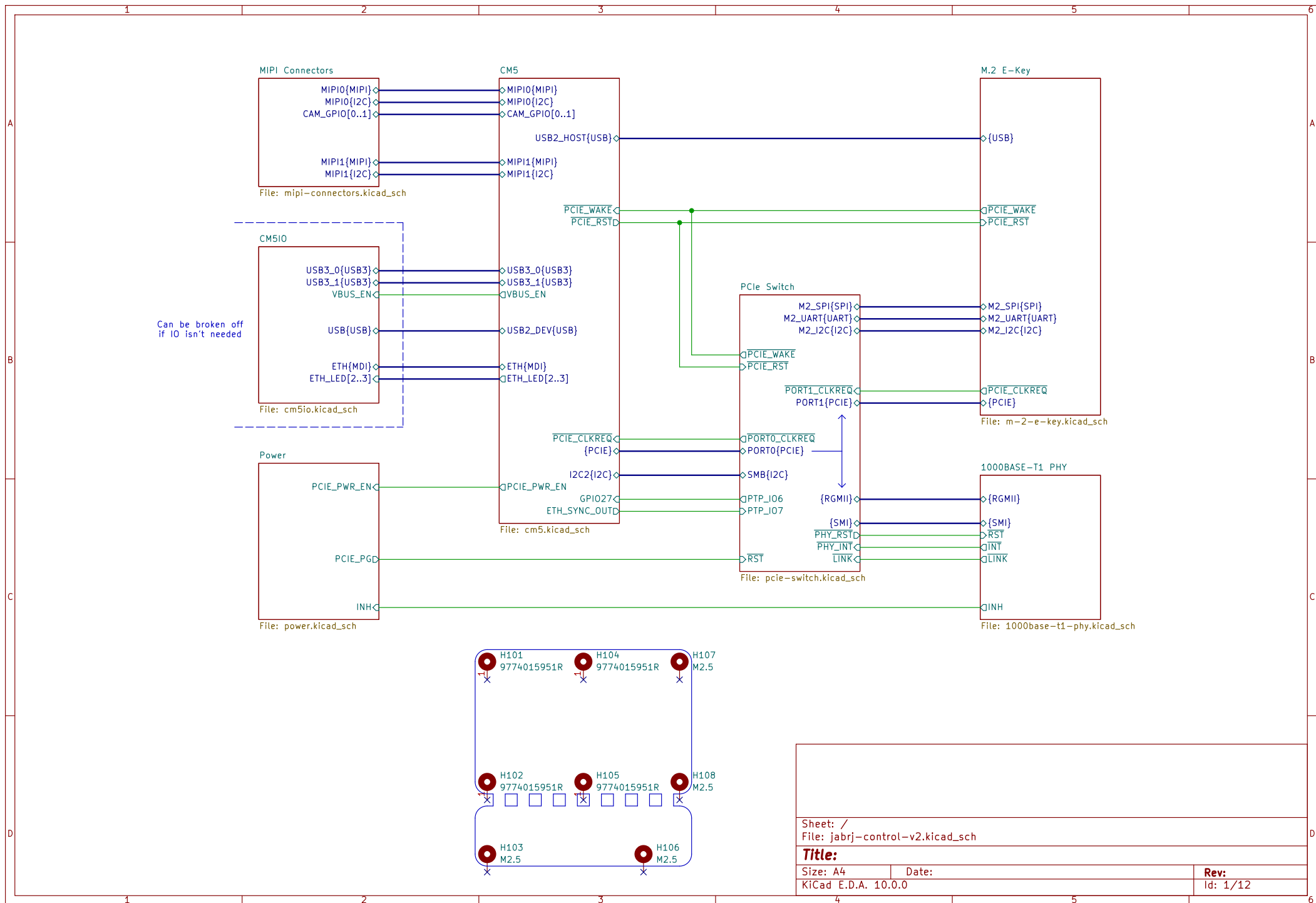




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File: jabrj-control-v2.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 1/12

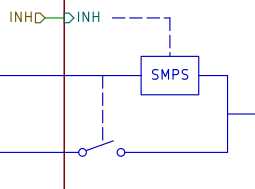
Power In



VIN (12V-24V)

<1750mA

5V Regulator



5V

CM5IO



VBUS

<4900mA

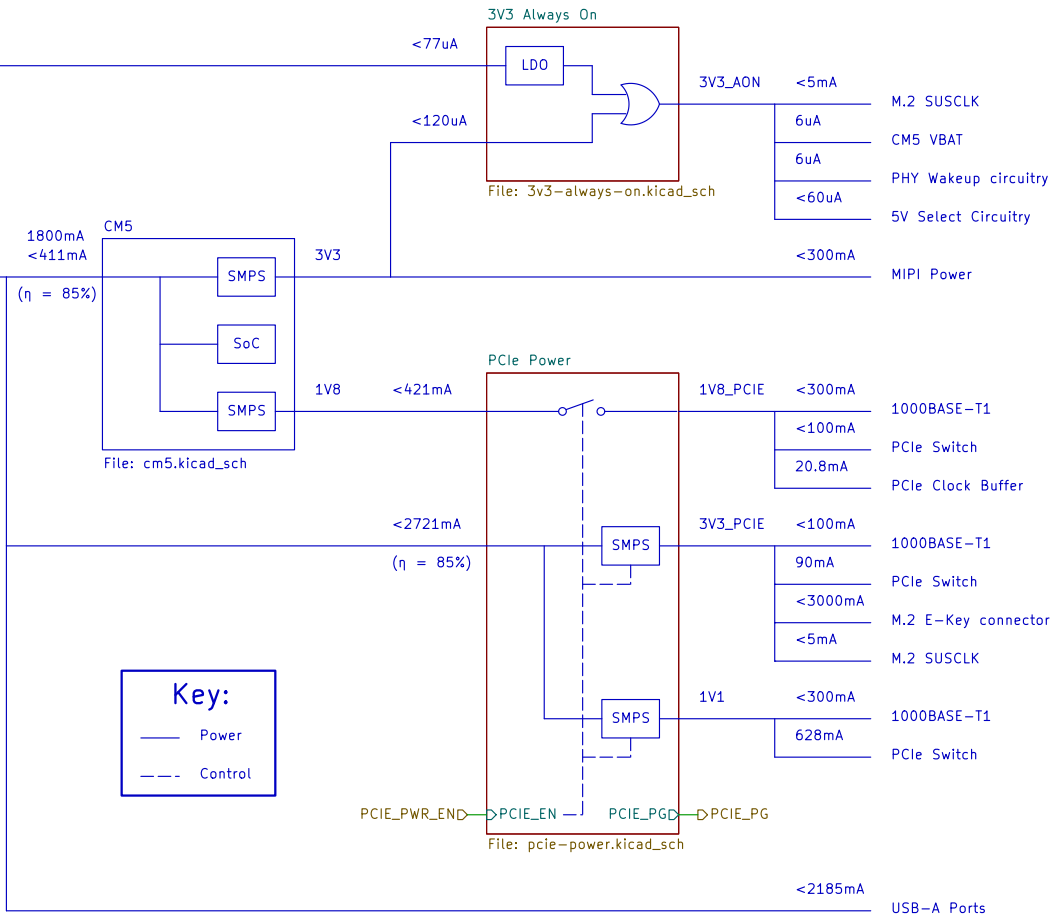
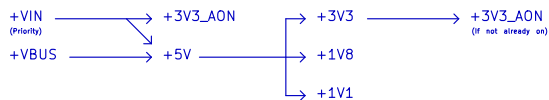
File: 5v-regulator.kicad_sch

The maximum power that can be consumed by the system is 13W, and the maximum power that could be consumed by peripherals (M.2 & USB-A ports) is 20.4W. Not all sources are able to provide this power so the peripheral power must be limited:

- 5V USB (3A capable): 2W for peripherals
- 5V USB (5A capable): 12W for peripherals
- 12V VIN: 5W for peripherals
- 24V VIN: 20W for peripherals

Note that the USB-A ports cannot consume more than 10W combined.

Power up sequence:



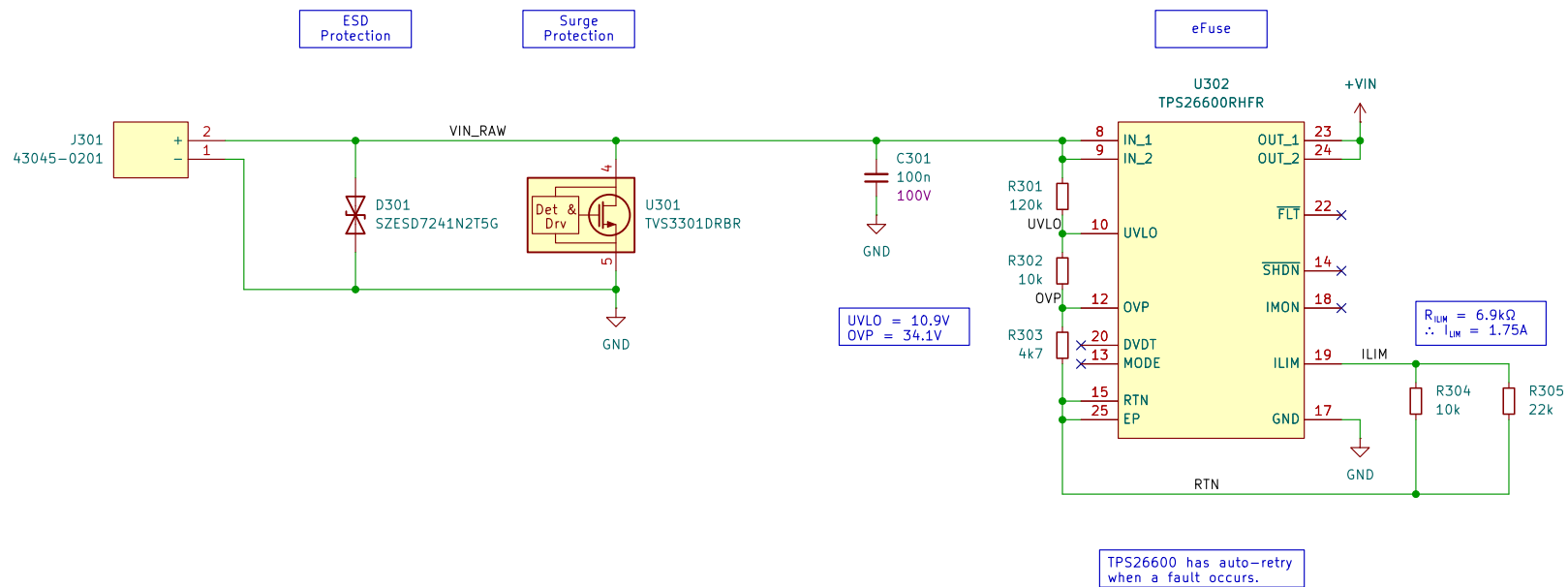
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File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 2/12



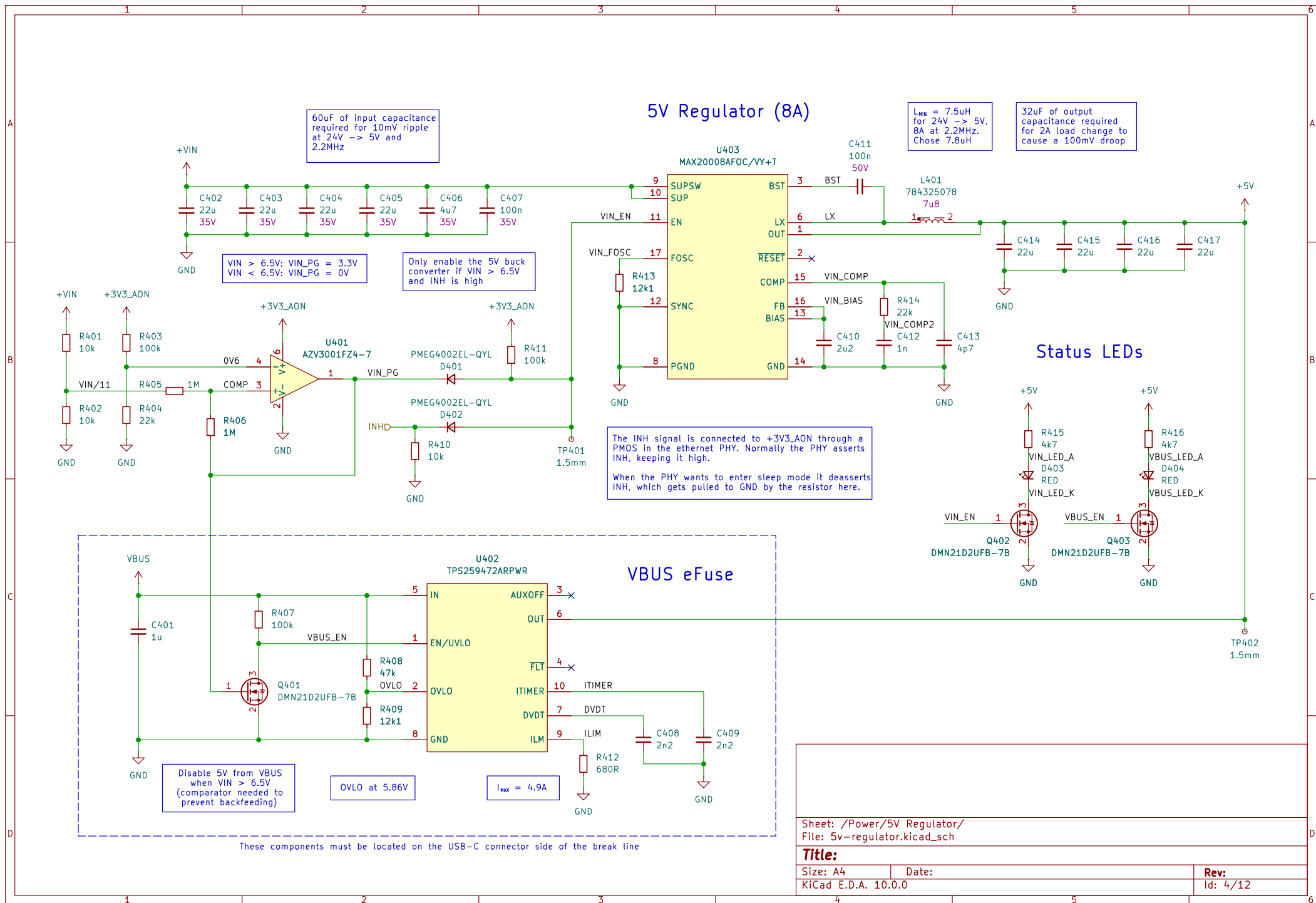
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File: power-in.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

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Id: 3/12



Sheet: /Power/5V Regulator/
File: 5v-regulator.kicad_sch

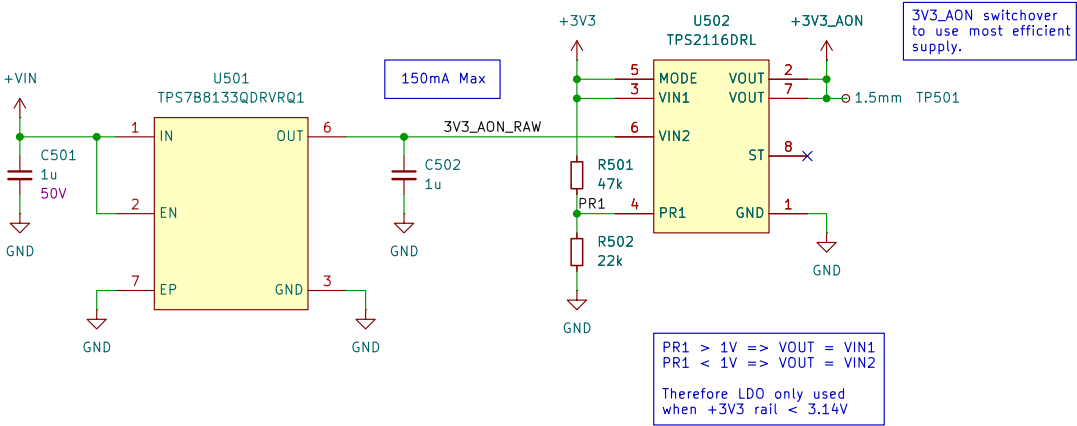
Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 4/12

3V3 Always on supply



Sheet: /Power/3V3 Always On/
File: 3v3-always-on.kicad_sch

Title:

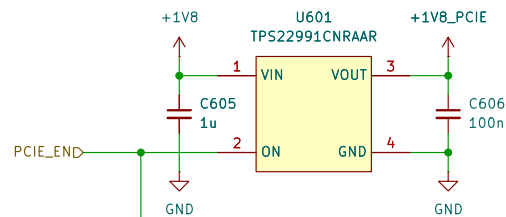
Size: A4

Date:

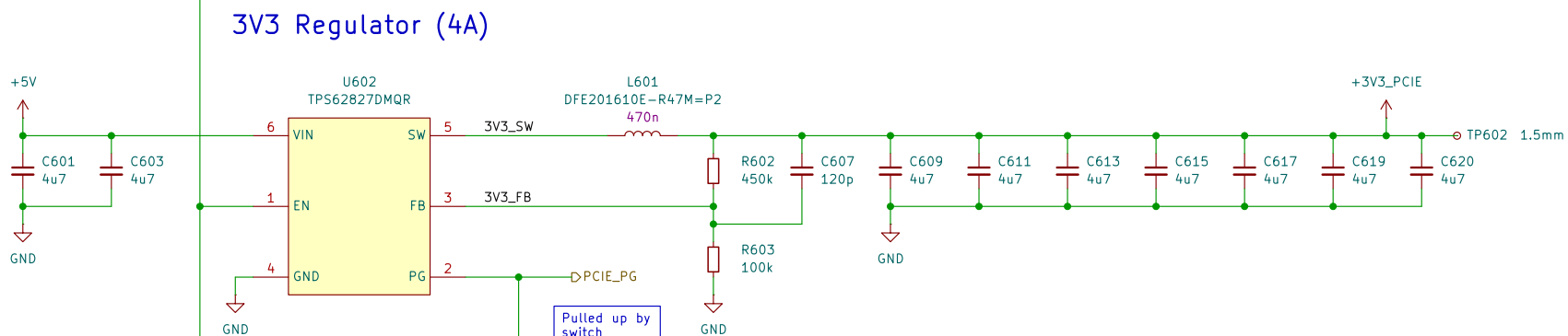
KiCad E.D.A. 10.0.0

Rev:

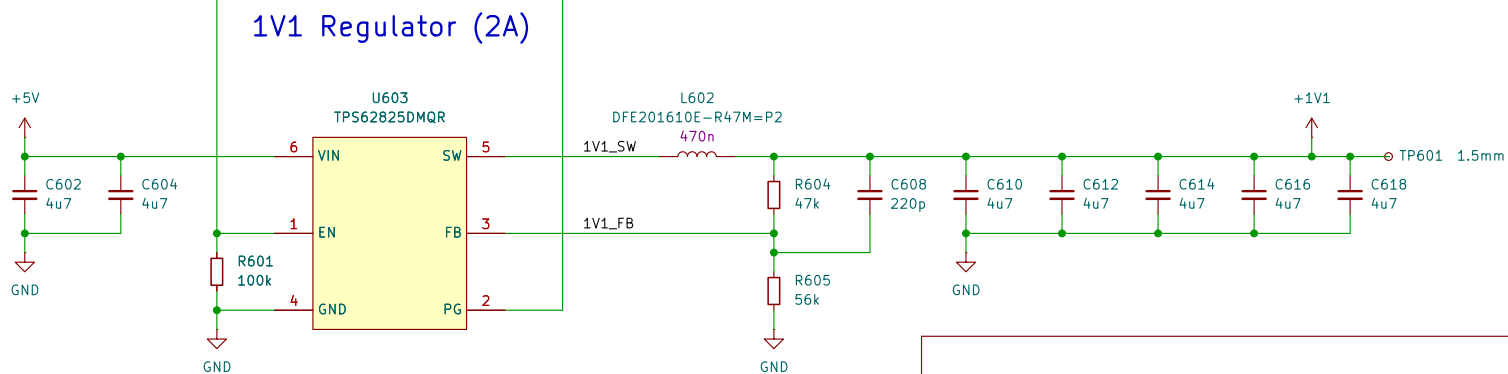
Id: 5/12



1V8 Load Switch (3A)



3V3 Regulator (4A)



1V1 Regulator (2A)

Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

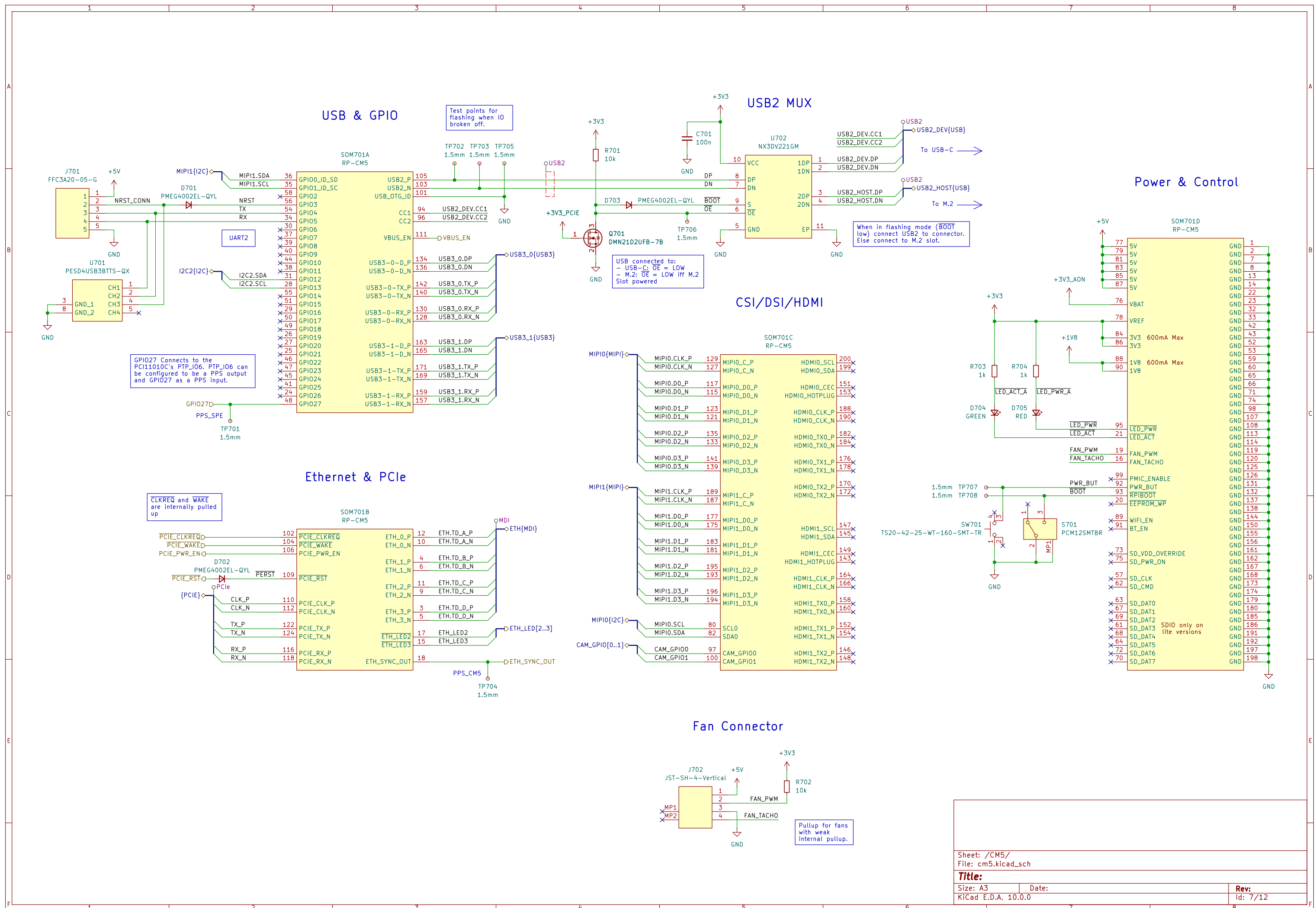
Sheet: /Power/PCIE Power/
File: pcie-power.kicad_sch

Title:

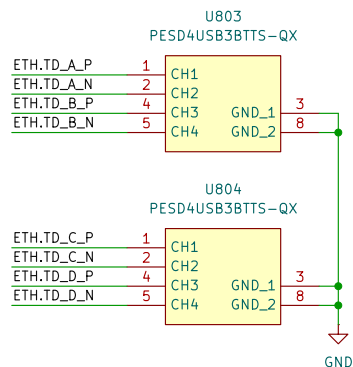
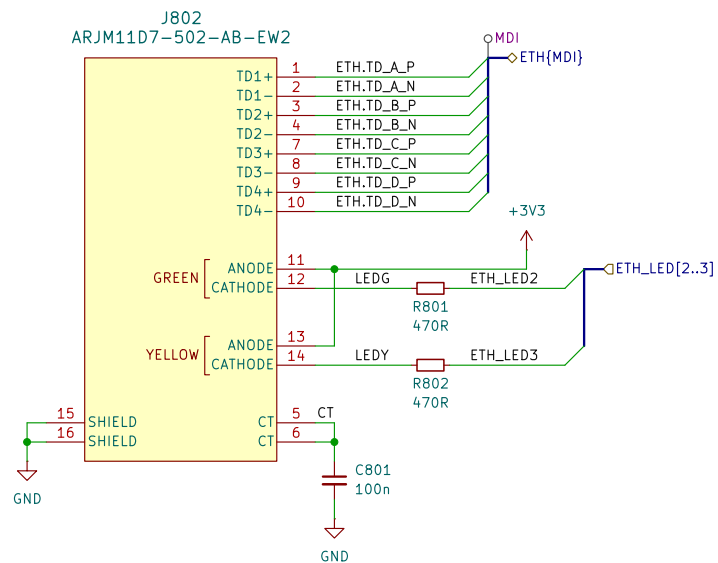
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KiCad E.D.A. 10.0.0

Date:

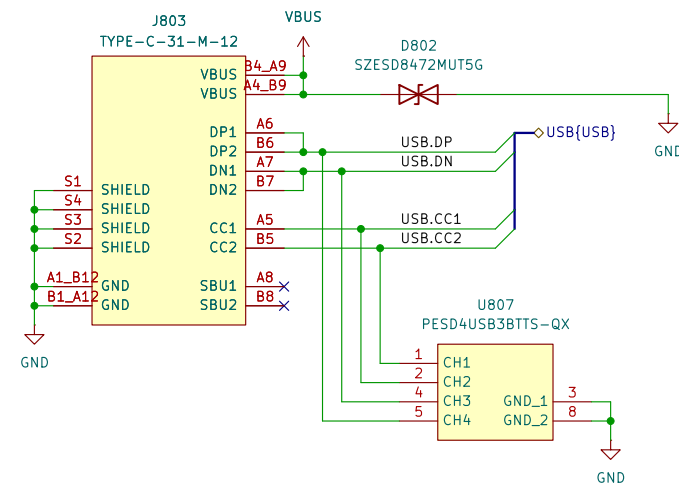
Rev:
Id: 6/12



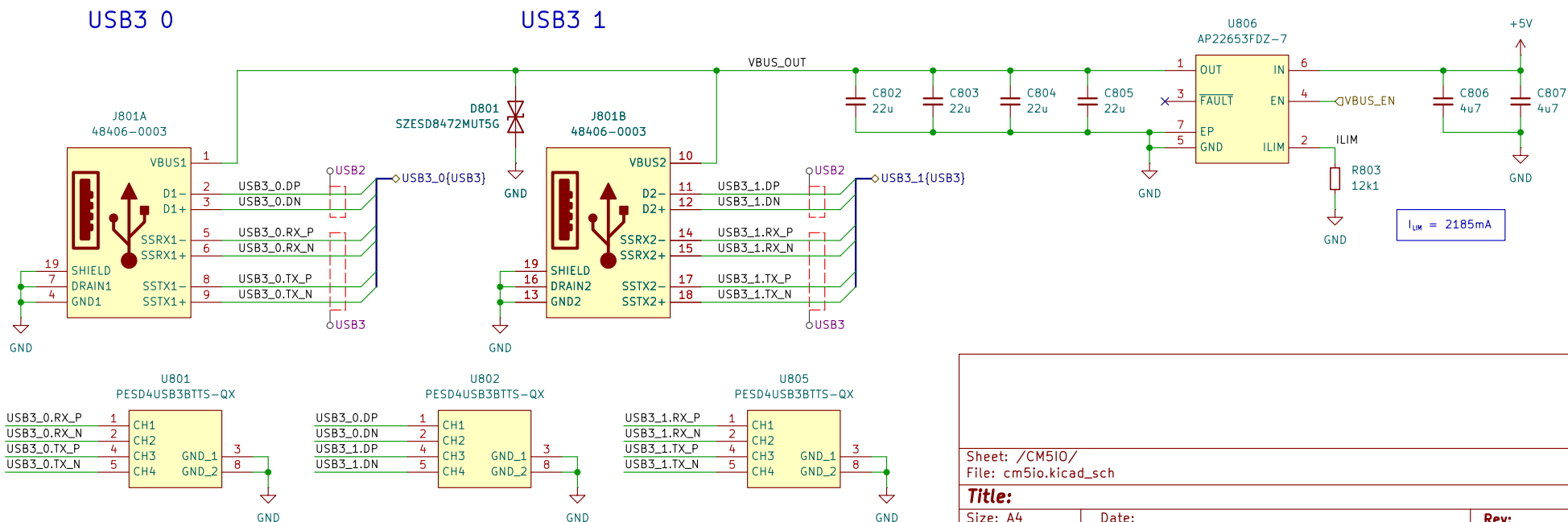
Ethernet RJ45



USB2



VBUS Load Switch



Sheet: /CM510/
File: cm510.kicad_sch

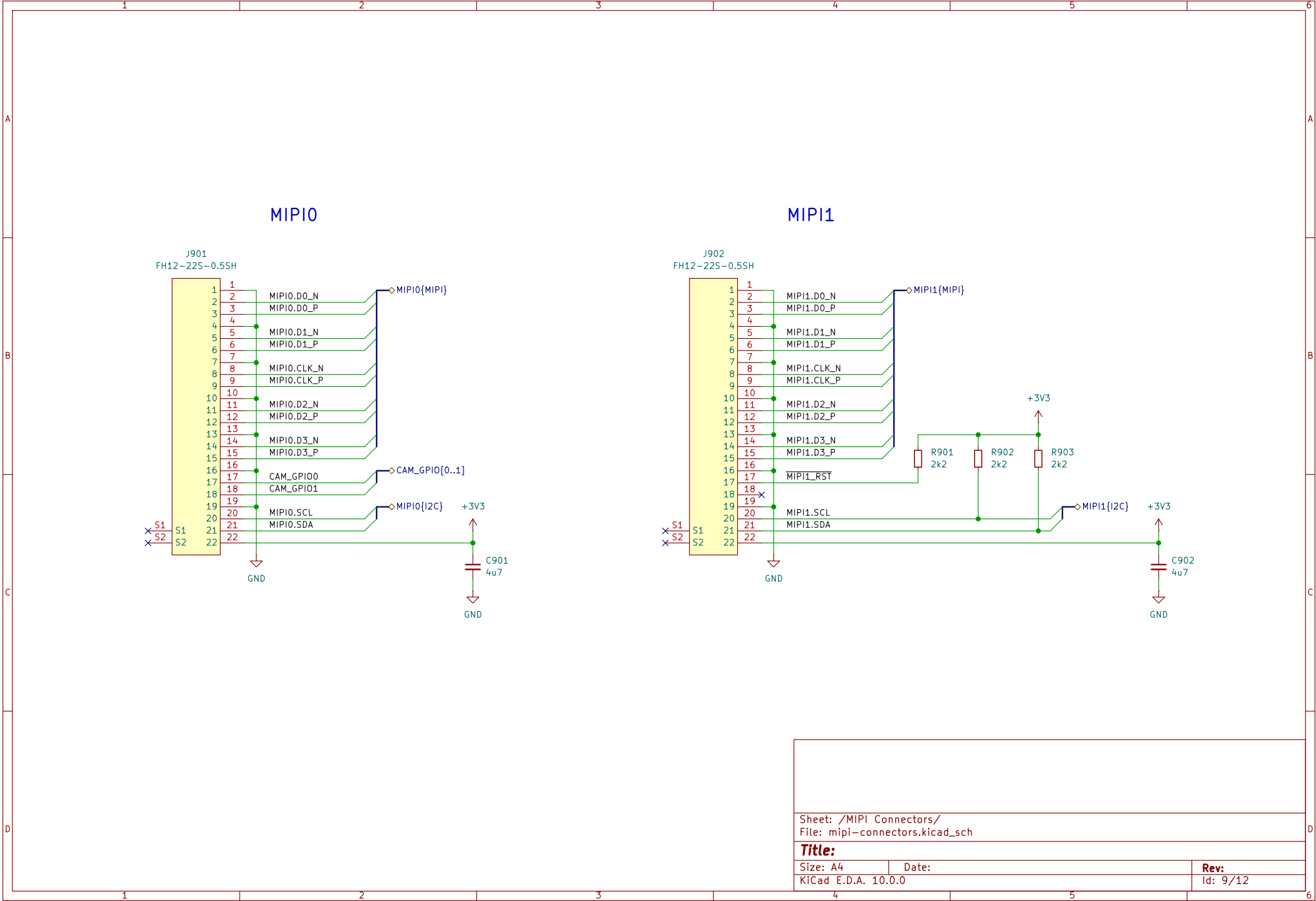
Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 8/12



Sheet: /MIPI Connectors/
File: mipi-connectors.kicad_sch

Title:

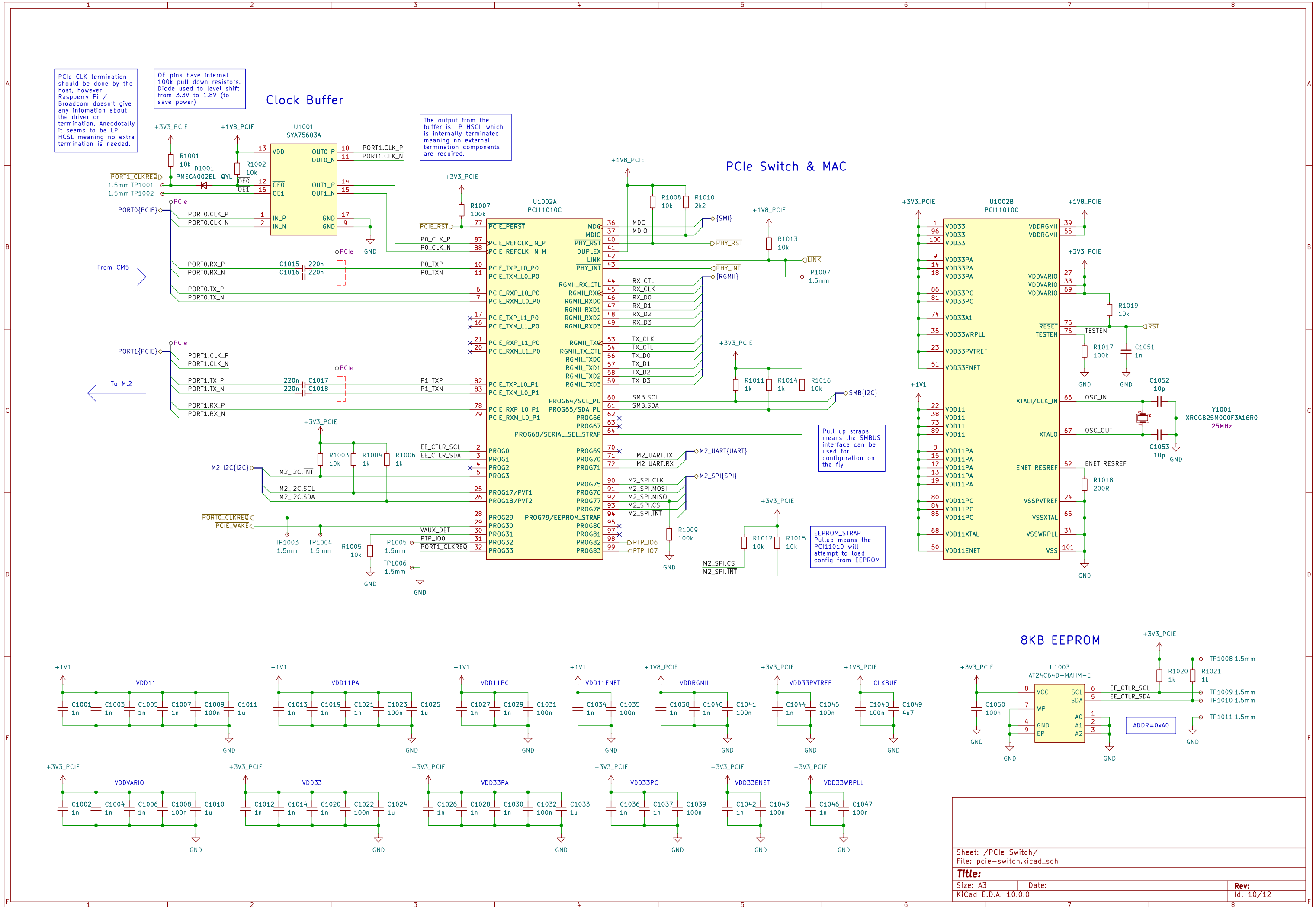
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Date:

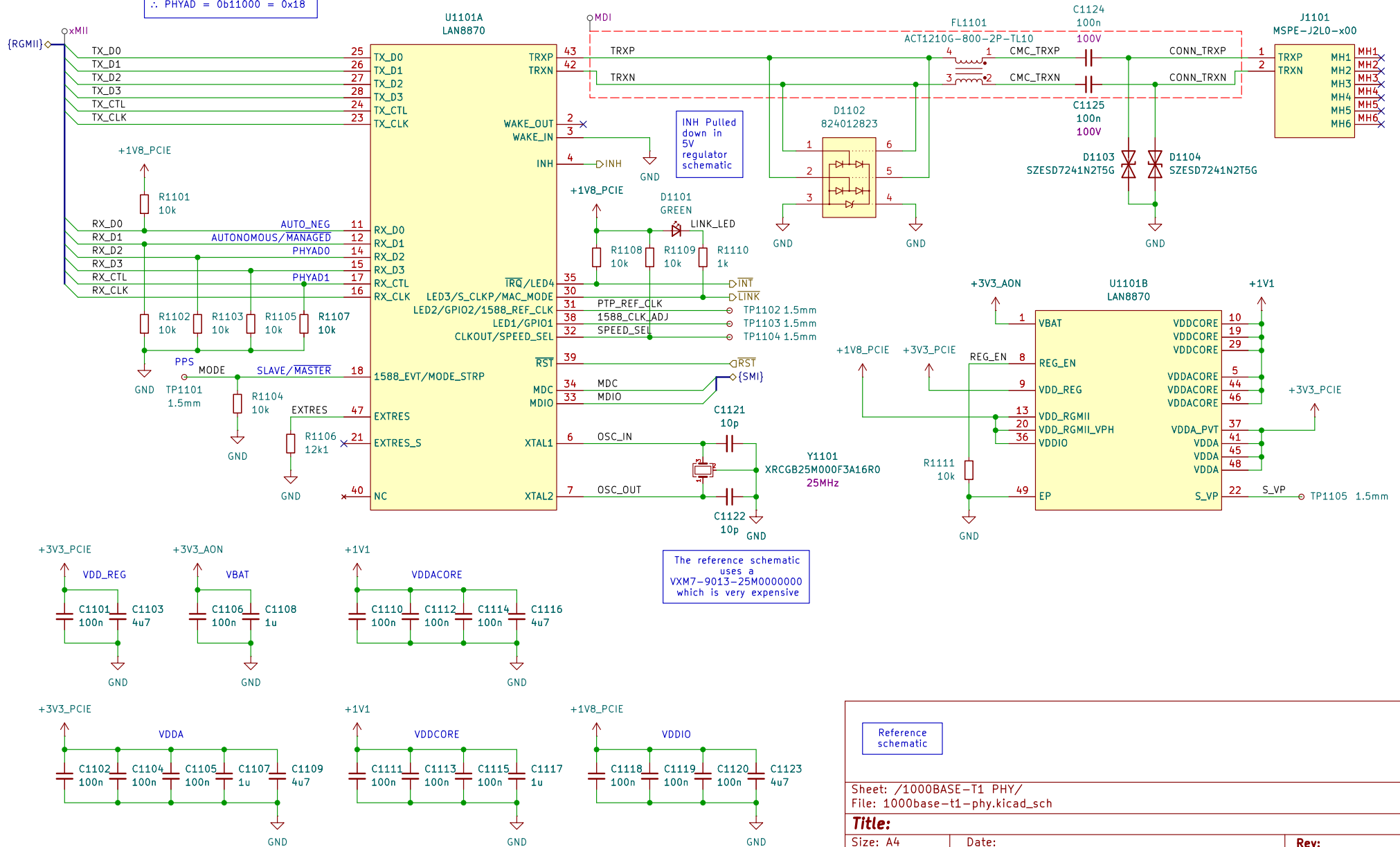
KiCad E.D.A. 10.0.0

Rev:

Id: 9/12



When in managed mode:
 - SPEED_SEL = PHYAD4
 - AUTO_NEG = PHYAD3
 - MODE = PHYAD2
 $\therefore$ PHYAD = 0b11000 = 0x18



Reference schematic

Sheet: /1000BASE-T1 PHY/
 File: 1000base-t1-phy.kicad_sch

Title:

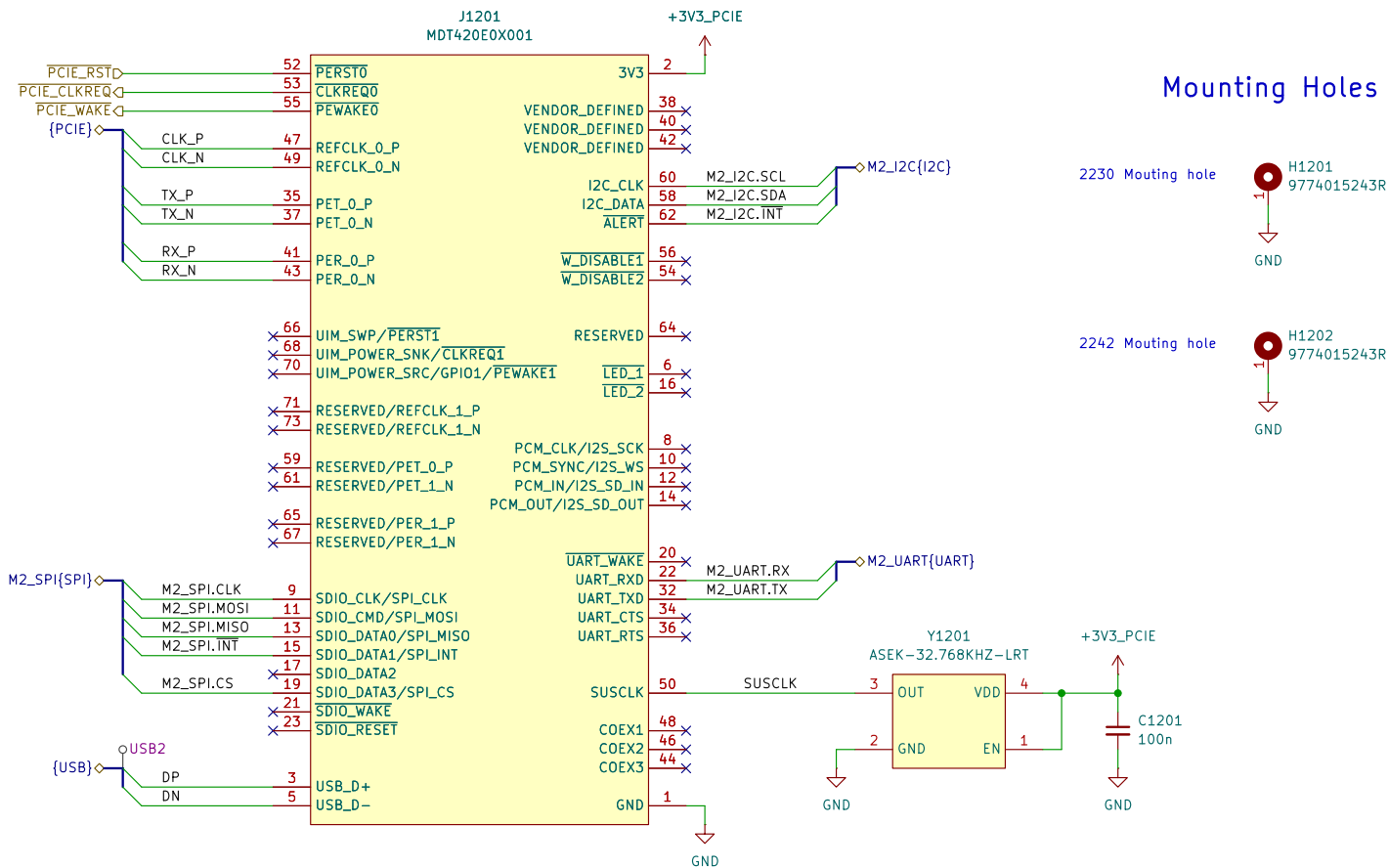
Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 11/12

M.2 E-Key



Sheet: /M.2 E-Key/
File: m-2-e-key.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:

Id: 12/12